2. Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

Aim:

To implement a monoalphabetic substitution cipher in which each plaintext letter maps to a unique ciphertext letter.

Procedure:

- Start the program.
- Read the plaintext.
- Use a substitution alphabet as the key.
- Replace each letter with its corresponding ciphertext letter.
- Display the encrypted text**.**

Program (C):

```
#include <stdio.h>
#include <ctype.h>

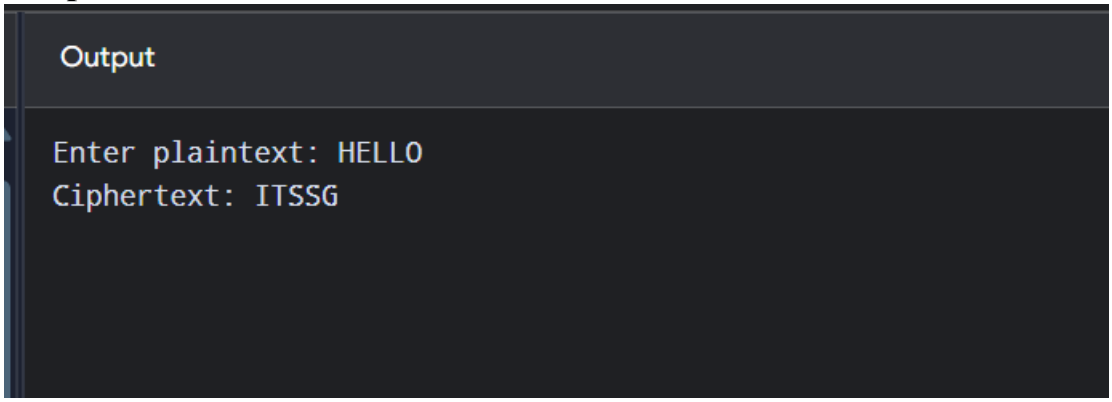
int main() {
    char s[100];
    char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
    int i, x;
    printf("Enter plaintext: ");
    fgets(s, sizeof(s), stdin);

    for (i = 0; s[i] != '\0'; i++) {
        if (isalpha((unsigned char)s[i])) {
            x = toupper((unsigned char)s[i]) - 'A';
            s[i] = key[x];
        }
    }
    printf("Ciphertext: %s", s);
    return 0;
}
```

Input:

HELLO

Output:



Result:

Thus, the monoalphabetic substitution cipher was implemented successfully.